

A standard Garden-of-Eden counterexample to conjecture 00000001245

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Definitions and claim

An elementary cellular automaton (ECA) is specified by a Wolfram rule code $r \in \{0, \dots, 255\}$. For a neighborhood $(\ell, c, t) \in \{0, 1\}^3$, the output is bit $4\ell + 2c + t$ of r . A configuration is a function $x : \mathbb{Z} \rightarrow \{0, 1\}$. The global map applies the local rule to $(x(i-1), x(i), x(i+1))$ at every integer i . A nonempty finite word is a Garden-of-Eden (GoE) word when it occurs in no global output, at any translated location. Its density is the proportion of its symbols equal to 1.

The conjecture asserts that the infimum of the minimum GoE-word densities over ECAs is $1/4$. If a rule has no GoE word, its minimum is undefined; we include only genuinely attained per-rule minima. The counterexample also applies if one takes infima within nonempty GoE-density sets instead.

Theorem 1. *Over all ECA rules with an attained minimum GoE-word density, the infimum of these minimum densities is 0, not $1/4$.*

Proof. Rule 255 has all eight local output bits equal to 1. Consequently its global image consists only of the all-1 configuration. The one-letter word 0 cannot occur in this image at any location: it is a genuine GoE word in the standard, unrestricted double-infinite sense. Its density is 0. Every finite-word density is nonnegative, so this word attains the minimum density for rule 255. The set of defined per-rule minima contains 0 and is bounded below by 0. Its infimum is therefore 0. $\square$

Correction of the prior closed submission

PR #174 incorrectly tested only periodic, length-two cyclic preimages. It claimed that the word 00 had no preimage under rule 133. Under the standard definition, the input patch 1011 has adjacent neighborhoods 101 and 011, both mapped to 0 by rule 133. Extend this patch by zeros outside its four sites to obtain a double-infinite preimage of 00. Thus that earlier witness was false. The separate observation that rule 51 is bijective addresses only the parenthetical candidate in the problem and does not disprove the claimed infimum. The present argument instead uses a standard GoE word of rule 255 to refute the full infimum statement.

Formal verification

The Lean project defines all 256 Wolfram rules, configurations on $\mathbb{Z}$, the local and global maps, translated finite-word preimages, GoE status, word density, attained minima, and their set-theoretic infimum. It proves the rule-255 truth table, the absence of a preimage for the one-letter zero word, nonnegativity, attainment at zero, and the exact infimum. It also verifies the actual rule-133 preimage described above. Lean 4.33.1 and a fixed Mathlib revision are pinned. The build has no unfinished proofs, native evaluation, or additional axioms; the printed dependencies are propositional extensionality, classical choice, and quotient soundness. Only the eight local neighborhoods are enumerated; no finite cyclic-boundary surrogate or long brute-force search is used.